

MPC-in-the-Loop Design Evaluation of a Hybrid Tensegrity Humanoid

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract—This paper presents a simulation-based design study of a full-size humanoid whose structure is a prestressed tensegrity network of struts and cables rather than a rigid skeleton. It addresses which joints of such a machine must remain conventional bearings for the robot to be controllable. Three simulated machines of identical anthropometry are compared under receding-horizon iLQG control: a pure tensegrity machine, a fully hinged 27-DoF tendon-driven machine, and a hybrid. A wrench-closure analysis shows that a bearing-free cable joint requires both overlapping cages and counter-wound cable families to resist disturbances. The pure machine stands when anchored but presents 210 free degrees of freedom, for which no controller was attempted. The fully hinged machine walks, but its tension network increases peak ground reaction in passive drops relative to the same skeleton without cables, and its cable transmission cannot deliver the specified joint torques at mid-stride poses. The hybrid retains bearings at the hips and knees, where torque demands are large and pose-dependent, and uses tensegrity joints at the ankles and waist, where impact loads enter the structure. At 1.68 m and 20.1 kg, with cables at the specified series-elastic stiffness, it walks at 0.20 m/s in 10/10 trials with 0.8 kg of articulation hardware, and a gait developed on a model with $46\times$ softer cables transfers across the 20–168 kN/m stiffness range without retuning. Rigidizing the tensegrity joints increases gait speed but removes the 27–32% impact attenuation measured at the ankle. The results indicate that the concept warrants hardware development: the machine has a third to two thirds of the mass of size-matched humanoids and is controllable by MPC.

I. INTRODUCTION

Mass is a primary constraint in humanoid design: it determines actuator torque and hence motor and battery mass, the energy of unintended collisions, and the hazard a human-sized machine poses to people nearby. Conventional humanoids, serial chains of rigid links with actuators at the joints, concentrate mass distally and couple strength to weight.

Tensegrity structures [1], [2] follow a different principle: struts are held apart by a network of pre-tensioned cables, every member is loaded axially, and stiffness is governed by prestress rather than by cross-section. These properties have motivated tensegrity landers and rolling robots [3]–[5] and tensegrity joints in sub-scale legs and spines [6]–[9], but not to a walking machine of human size, where their advantages cannot be settled by static analysis.

This paper reports one iteration of a design loop in which model-predictive control (MPC) is the inner step: a parametric tensegrity kinematics is generated, its controllability under tension-only actuation is tested, actuators and members are sized from simulated load traces, and a receding-horizon iLQG controller [10]–[12] is used to balance and walk the machine.

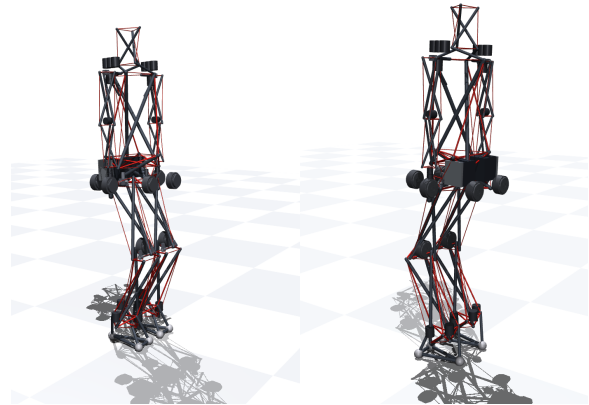


Fig. 1. The hybrid humanoid, front and rear views (1.68 m, 20.1 kg). Each segment is a three-strut prism cage; cables are shown in red. The hips and knees are bearings driven by AK70-class motors on the pelvis girdle and thigh cuffs. The feet and torso are free bodies held only by counter-wound cable interfaces (Fig. 3); XM540-class tension drives are mounted on the shank rings and torso. Two battery packs are carried on posterior pelvis rails.

We refer to this procedure as *MPC-in-the-loop design evaluation*; the design space is not searched automatically, and each structural change is justified by a measured change in the gait. A *fully hinged* 27-DoF tendon-driven machine walked first, but its cable transmission fell short of the required torque at 13 of 27 DoF over the gait (Sec. IV) and its cable network increased impact loads (Sec. V). A *pure tensegrity* machine without hinges collapsed until its joints were redesigned with overlapping cages and counter-wound cable families (Sec. VI), and only this bearing-free joint attenuated impacts. The *hybrid* combines the two (Sec. VII), and a joint-count study over its rigidized variants quantifies the trade-off (Sec. VIII).

The purpose of the study is to determine whether the concept justifies hardware development, i.e., whether a tensegrity humanoid can be realized at a fraction of conventional mass while remaining controllable by a predictive planner. Both conditions are met (Table I, Sec. VII). The gait was developed with interface cables at 2.4 kN/m, approximately $46\times$ softer than the specified 20–168 kN/m series-elastic element, because this is the stiffest cable the planner integrates at a step fast enough for design iteration (Sec. III). All headline results are obtained at 40 kN/m, within the specified range; results that remain at the *development configuration* are labeled as such, and the geometric results are independent of cable stiffness.

The contributions are: (i) a 1.68 m, 20.1 kg hybrid humanoid, specified to tube, rope and drive class, that walks

at 0.20 m/s in 10/10 trials under receding-horizon iLQG at the specified cable stiffness, compared with 30–80 kg for built machines of similar stature (Table I); (ii) a bearing-free joint between stacked prism cages and the rule that classical theory implies for it: overlap provides force closure, but only a counter-wound cable pair provides the strict self-stress [13], equivalently wrench closure [14], [15], that tension-only actuators need to reject disturbances; (iii) a pose-dependent authority bound, the per-DoF form of the wrench-feasible workspace [16], applied as a design gate over the whole gait; and (iv) a joint-count study over rigidized variants of one machine, yielding measured trade-offs and an allocation rule. To our knowledge, no prior work compares bearings against cable interfaces within one legged machine. The MPC architecture and the pipeline that regenerates every reported quantity from logged runs are supporting infrastructure.

II. RELATED WORK

Tensegrity robotics. Robotic tensegrity [1], [2], [17] has been demonstrated primarily in rolling machines such as SUPERball [3], [4]; see [5] for a survey. The closest control precedent is MPC with inverse-statics tension distribution on tensegrity spines [18], on low-DoF, quasi-statically loaded modules. The ULTRA Spine [19] and Laika [20] place a tensegrity spine between rigid legs; the present design instead places the tensegrity interfaces at the joints through which ground impacts enter. Bearing-free tensegrity joints have been built at limb scale: elbows and shoulders [21], a 3-DoF joint with wrench closure enforced at every equilibrium [15], knee- and hip-inspired flexural joints [6], a one-third-scale rigid-tensegrity biped knee demonstrated in hardware [7], a leg whose tensegrity joints reduce impacts by about one third [8], and variable-stiffness and humanoid spines [9], [22]. None addresses which joints of a complete humanoid should be tensegrity and which bearings. Classical results (strict self-stress [13], the elastic-geometric stiffness split [23], form-finding [24]) are applied without extension.

Tendon-driven and compliant humanoids. Kenshiro [25] and Kengoro [26] route about one hundred muscles over a human-mimetic skeleton under the same unilateral tension constraint; here the tendons constitute the structure itself. Series-elastic actuation [27], [28] reappears as a requirement on the tension members, and soft exosuits [29] show that textile load paths carry such forces, motivating an apparel-mounted structure as a longer-term goal.

Co-design and MPC. Morphology and control have been optimized jointly by optimal control [30] and reinforcement learning [31]; the present work reports one manual iteration with a predictive controller as the inner step. iLQG synthesizes whole-body humanoid behavior [11], [32], whole-body MPC has run on torque-controlled humanoids [33], [34], and MuJoCo MPC [12] provides the planners used here. The gait reference follows Raibert [35] and the capture point [36]; torque budgets follow Winter [37].

Table I shows a nearly threefold spread in mass at fixed stature, indicating that architecture rather than scale is the

TABLE I
HUMANOIDS OF COMPARABLE STATURE. THE SIMULATED ROWS ARE MASS BUDGETS WITHOUT WIRING, SKIN OR SENSORS (LOWER BOUNDS). THE HYBRID’S 46 DOF COMPRISE THE FLOATING BASE, 22 HINGES AND THREE FREE BODIES (TWO FEET, TORSO), DRIVEN BY 30 ACTUATORS.

robot	m	kg	kg/m	DoF	actuation
this work: hybrid (sim)	1.68	20.1	11.9	46	hinge+tensegrity
this work: hinged (sim)	1.65	27.7	16.8	27	tendon, hinged
1X NEO Gamma [38]	1.65	30	18.2	—	tendon, soft shell
HRP-4 [39]	1.51	39	25.8	34	rigid, geared
Unitree G1 [40]	1.32	35	26.5	23	rigid, quasi-direct
Kengoro [26]	1.70	56	32.9	114	musculoskeletal
Fourier GR-1 [41]	1.65	55	33.3	40	rigid, geared
PAL REEM-C [42]	1.65	80	48.5	44	rigid, geared

dominant factor; the lightest built machines are tendon-driven with soft shells. The capability gap is equally significant: GR-1 walks at 1.4 m/s, whereas the hybrid reaches 0.20 m/s in simulation. Speed was not an objective of this study, which asks only whether the structure can be controlled at all with a standard MPC approach.

III. THE HYBRID DESIGN

The final design (Fig. 1) is 1.68 m from sole to crown and 20.1 kg, with 0.40 m thigh and shank segments (Winter anthropometry [37]). *The hips and knees are bearings*, because their torque limits, 40–80 N m, are the largest in the machine and are exercised through large excursions: each knee is an alloy-steel clevis pin in a 61802-class bearing pair (142 g) and each hip a three-axis gimbal (257 g), sized at a safety factor of 2.3 against the measured 1454 N stumble reaction. Articulation hardware at the hips and knees totals 0.8 kg, and the arm hinges add 0.12 kg of shoulder gimbal hardware. AK70-class drives on the girdle and thigh cuffs actuate these hinges through tendon loops at a reduction of approximately 3:1 (12 mm spool, ~ 37 mm joint radius), as in the hinged machine. *The ankles and waist are tensegrity joints*. Each foot is a tri-pad plate (heel and two forefoot corners, following the insertion geometry of the biological ankle [43]) whose mast overlaps into the shank cage, and the torso is a free body whose cage nests into the pelvis girdle. Both are held only by the counter-wound cable interfaces of Fig. 3, with twelve cables each, no bearing, and force and wrench closure by construction. Because these cables act at the cage radius rather than through a ~ 37 mm moment arm, the same moment requires four to five times less tension, and 0.165 kg XM540-class drives replace 0.96 kg units. Each arm has seven DoF (shoulder gimbal, elbow, three-axis wrist) with all drives modeled; during walking only the two swing DoF per arm are planned, the others holding position on servo stiffness. The neck is rigid, and two 2.5 kg battery packs are carried on posterior pelvis rails.

Every segment is a Snelson three-strut prism at its form-found equilibrium twist: released free-floating in zero gravity, the cell converges to $29.8^\circ \pm 0.4^\circ$, consistent with the theoretical 30° [24]. Two earlier variants of the same anthropometry bound the design space: the *fully hinged* 27-DoF machine, in which every joint is a tendon-driven bearing, and the *pure*

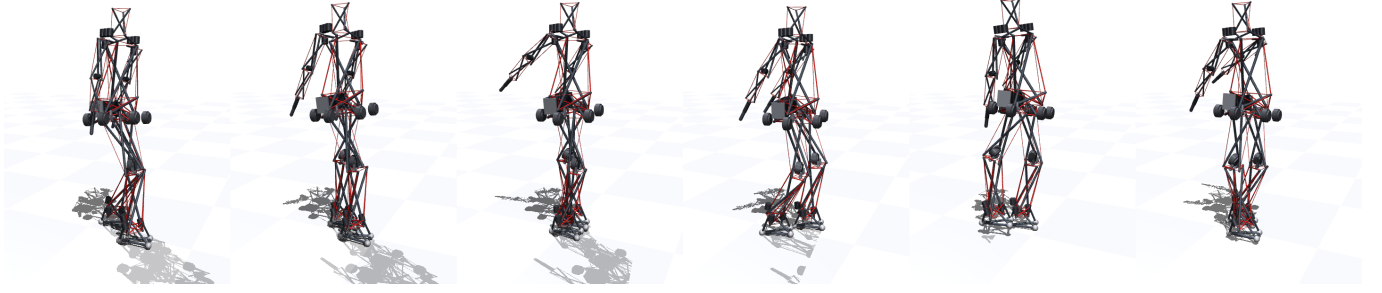


Fig. 2. The hybrid walking under iLQG in the stride-lengthened configuration of Sec. VII (0.23 m/s, 10/10 trials). Frames are 7/6 of a gait cycle apart, so that the displayed phase advances by one sixth of a cycle per frame. Struts are dark gray and tension members red; the feet and torso are free bodies. The arms hold a servo-referenced posture. Development configuration (interface cables at 2.4 kN/m). See the accompanying video.

TABLE II
MASS BUDGET OF THE HYBRID. ALL DRIVES ARE MOUNTED PROXIMALLY, AND THE TENSEGRITY JOINTS CARRY NO BEARINGS.

item	n	kg	where
hip drives, AK70-class	6	3.12	pelvis girdle
knee drives, AK70-class	2	1.04	thigh cuffs
ankle tension drives, XM540	6	0.99	shank rings
waist tension drives, XM540 (one per family)	2	0.33	torso ring
arm drives, XM540/XM430	14	1.65	torso + arm
articulation hardware (gimbals, clevises)		0.80	hips, knees
shoulder gimbal hardware		0.12	shoulders
battery packs (incl. rails), hot-swappable	2	5.10	pelvis rails
compute + power		0.50	pelvis
structure: cages, cables, feet, masts		6.42	distributed
total (compiled model)		20.1	

tensegrity machine without bearings (Sec. VI). Joint torque limits follow human walking moments scaled to robot mass at a safety factor of about 2 (80 N·m at hip pitch and knee, 70 N·m at the ankle, 30 N·m at the hinged shoulder, 15 N·m XM540-class on the hybrid).

Member specification. Each member is dimensioned from the simulated load cases rather than covered by a bulk allowance (Table II). The 60 CFRP tubes (12–20 mm outer diameter) are governed by the ground reaction entering a cage through its joint (485 N per strut in a 0.1 m stumble); Euler sizing at a safety factor of 2 nevertheless places every strut at the minimum manufacturable wall (0.5 mm, worst margin 6.0×), so the compression network (0.67 kg of tube, 0.36 kg of fittings) is limited by manufacturability rather than load. Tendons are 12-strand UHMWPE rope of 1–4 mm diameter, sized from the larger of the gait’s 99th-percentile tension and $\tau_{\max}/r_{\text{eff}}$ (worst RMS utilization 12%). The rope has a mass of 0.10 kg, whereas its 444 terminations have a mass of 1.78 kg and the 93 node clusters 0.84 kg; the mass of such a network is set by its number of interfaces rather than its number of cables.

Cable stiffness: model versus hardware. The specified rope has a stiffness of 0.3–7.1 MN/m. A deliberate series-elastic element of 20–168 kN/m [27] is required so that pre-stress can be set at assembly, and the transmission analysis assumes it. The hybrid was developed with interface cables at 2.4 kN/m, the *development configuration*. This value was

chosen for the planner rather than the hardware: it is the stiffest interface that the 8 ms planner step integrates stably (mode frequency times step 1.84, against a limit of 2), and at this step the planner runs approximately $8\times$ faster than at the design point (0.29 versus $0.034\times$ real time), which made the tuning runs of the design iteration tractable. Once the design was fixed, the walker was rerun with smaller steps at the 40 kN/m design point and across the specified range (Table IV); the joint testbed of Sec. VI is also simulated at 40 kN/m. The hinged machine’s experiments use 600–2500 N/m, with the impact and stiffness experiments of Sec. V rerun at physical stiffness.

IV. MODELING, AUTHORITY, AND CONTROL

The machines are modeled in MuJoCo [44] by a parametric generator that produces, from a single description of the node geometry, a *joint-torque* model (motors at the hinges; the planner’s model) and a *cable-driven* model (114 tension-only actuators). Cables must be tension-only: a MuJoCo tendon with a single spring length exerts a compressive force when shortened, so all cables use a deadband spring $F = k \max(0, \ell - \ell_0)$ with ℓ_0 set by the prestress fraction and recomputed from the compiled model. Because a cable helixed around a limb has a moment arm of only 16–25 mm about the limb axis, the yaw DoF carry coaxial capstan drums of 35–55 mm diameter.

Cable authority. With tension-only actuation, the achievable torque about DoF j is bounded by

$$\tau_j^+ = \sum_c \max(0, M_{jc}) \bar{F}_c, \quad \tau_j^- = \sum_c \min(0, M_{jc}) \bar{F}_c, \quad (1)$$

where $M_{jc} = \partial \ell_c / \partial q_j$ is the moment arm of cable c about joint j and $\bar{F}_c$ its tension limit. This is the per-DoF projection of the wrench-feasibility condition for cable-driven parallel robots [14], [16]; no novelty is claimed for the bound, only for its application as a requirement over the entire trajectory. Applied in this way, the bound identified three geometric defects in an early iteration of the design, among them zero authority in one direction at the hip yaw and wrist, because all cable families shared one handedness (Sec. VI). Evaluated at the assembly pose alone, the bound overstates the available authority (Table III): at assembly all 27 DoF are bidirectional

TABLE III
CABLE AUTHORITY (EQ. 1) FOR THE FULLY HINGED VARIANT: NUMBER OF DOF SHORT OF THEIR TORQUE LIMIT AND WORST MARGIN, BY EVALUATION POSE AND ACTUATION SET.

configuration	actuation set	DoF short	worst margin
assembly pose	all cables	0/27	1.01×
gait poses	all cables	10/27	0.00×
assembly pose	36 loops	6/27	0.66×
gait poses	36 loops	13/27	0.00×

(worst margin 1.01×), whereas at logged gait poses 10 of 27 DoF fall below their torque limit (hip pitch 0.87×, knee 0.61×), and the 72-cable, 36-motor bill of materials leaves 13 of 27 short. The exact zeros in Table III correspond to capstan drums whose wrap disengages at certain wrist and shoulder angles. Delivering 80 N·m at the leg cables' 1.5 kN limit requires $r_{\text{eff}} \geq 53.3$ mm, whereas the cage provides 37.5 mm at hip pitch. This shortfall is the quantitative reason for retaining bearings at the hips and knees; a second cable of the same family loses its moment arm at the same poses.

Predictive control. The controller is the asynchronous iLQG planner of MJPC [11], [12] with control limits [32]; the horizon and planning step are 0.6 s and 20 ms for the hinged task and 0.42 s and 8 ms for the hybrid (set by the ankle-cable foot mode), with the plant integrated at 4 ms. Rather than shaping the cost by hand, which yields stepping in place, the controller tracks an *analytic gait reference evaluated inside the cost*: gait phase is a function of time alone, the pelvis tracks a lateral sway that unloads each foot before it lifts, the swing feet follow a sinusoidal clearance profile, and the landing point is biased by the center-of-mass velocity error following Raibert [35] and the capture point [36]; flat-foot inverse kinematics converts foot targets into joint references. A trial counts as a walk if the pelvis remains above 0.65 m and the robot travels more than 1 m between 3 and 15 s (the hinged machine's suite uses pelvis height above 0.55 m only); upright but stalled trials are reported separately.

Walking results, fully hinged variant. This machine validates the pipeline and provides the load traces used for dimensioning: it walks at 0.21 ± 0.01 m/s in 7/10 of ten trials, with a longest run of 6.1 m in 30 s, at $0.60\times$ real time. Drives are dimensioned from the tension traces and the push-recovery bandwidth requirement ($f_{\text{bw}} = \frac{1}{2\pi} \sqrt{k_{\text{ser}}/m_{\text{eff}}}$, $m_{\text{eff}} = I_r G^2/r^2$), yielding spools where stroke dominates (hips, knees, waist) and lever arms where bandwidth dominates (ankles, 46 Hz). This analysis also exposes a limitation of the fully hinged variant: its 36-motor bill of materials cannot meet the torque specification, since a single loop delivers 56 N·m at hip pitch against a requirement of 80 N·m, and Table III shows the shortfall increasing over the gait. Because the limiting quantity is the moment arm r_{eff} rather than the number of motors, the hybrid addresses it by retaining bearings at the hips and knees (Sec. III) rather than by adding actuators.

V. EVALUATION OF THE FULLY HINGED VARIANT

Five experiments were performed on the fully hinged variant. E1 and E2 compare it against a *rigid-equivalent baseline*,

the same 27-DoF model with the tension network removed and actuation held constant; E3–E5 examine the cable network itself.

E1, impact. In matched passive drops from 0.05–1 m, the tension network increases peak ground reaction at every height, by ++14% at 0.1 m and by +110–+197% at physical cable stiffness: a bearing places the elastic network in parallel with a stiff load path, which carries the impulse. **E2, stiffness.** Under a 200 mN differential endpoint probe, cables raise lateral stiffness by $2.4\times$, but sweeping assembly prestress from 0 to $5\times$ nominal changes it only within 11.6–16.6 N/m (by -2% at physical stiffness once slack is taken up), whereas shoulder co-contraction changes it from 14 to 115 N/m ($8.4\times$) without additional hardware. **E3, degradation.** All 15/15 tested single-cable cuts and all random losses of up to $k = 8$ cables survive standing plus a push; only an adversary selecting against the authority margin causes a fall at $k = 8$. No tested single cable is structurally critical, whereas one failed gearbox disables a joint of a rigid robot. **E4, proximal mass.** Relocating the motor masses to the joints they drive, at constant total mass, reduces speed from 0.21 to 0.17 m/s (success rate 7/10 versus 6/10, within sampling uncertainty). **E5, tendon realizability.** When the logged walking torques are mapped through the bounded tension distribution, a cable is pinned at its 1.5 kN limit in 10.9% of states, leaving up to 24 N·m undelivered, and the 99th-percentile demand is 2407 N, $1.6\times$ the rating. Re-planning within the achievable envelope, a gait exists for the full cable network (7/10) but not for the 36-motor bill of materials (0/10), which isolates the shortfall to the hinged variant's actuation set rather than to its cable topology.

VI. THE BEARING-FREE JOINT AND THE PURE TENSEGRITY VARIANT

Whether the hinged variant's lack of impact tolerance and prestress-tunable stiffness is a property of tensegrity or of its hinges is settled by an audit of its joints: every articulated interface constrains all three relative translations, and the wrench-closure linear program (closure holds if and only if $\max\{t : W\lambda = 0, \lambda \geq t\mathbf{1}\} > 0$ for the matrix W of unit-tension cable wrenches [14], [16]) finds 0/14 joints able to resist even pure forces by cable action alone, since every crossing cable pulls the segments together. Without hinges the robot telescopes and collapses, the pelvis falling below $0.6\times$ its home height within 0.6 s.

Force closure requires overlap; wrench closure requires counter-winding. In a tensegrity tower, successive stages overlap: the distal ring sits inside the proximal cage, giving the interface both a downward and an upward cable family (Fig. 3). On an isolated two-cage testbed (fixed proximal cage, three floating distal struts, 1 kg payload), force closure appears at an overlap of $0.1H$, and from $0.2H$ the joint also carries its design load. Force closure alone is insufficient: a margin $t_6 = 0$ means that the cable set has no strict self-stress, the classical existence condition for a tensegrity [13], hence no co-contraction state and no moment authority about some axis, so that tension-only actuators cannot oppose a disturbance

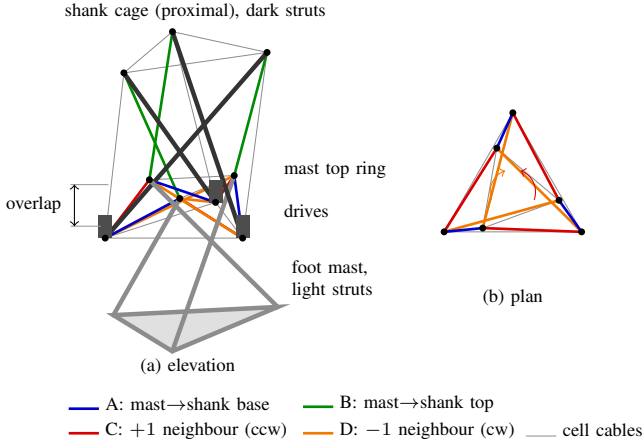


Fig. 3. The bearing-free ankle. (a) The three-strut mast of the foot extends $0.2H$ into the shank cage; twelve cables in four families of three connect the top ring of the mast to the shank. Families A and B provide force closure (the joint resists both separation and approach); C and D are the same cross family wound in opposite senses. (b) In plan view, C and D form opposite pinwheels. Either family alone leaves $t_6 = 0$, i.e., no self-stress state and no moment authority about some axis; together they close the wrench set at the same cable count. A and B are actuated (XM540 on 8 mm spools, approximately 250 N); C and D carry passive prestress. The waist uses the same interface.

about that axis. A controllable joint requires $t_6 > 0$. A single cross family wound in one sense cannot provide this; adding the counter-wound mirror family closes the wrench set at the same cable count, which is the general form of the opposite-handedness rule found for the yaw drums and the reason Snelson’s towers alternate the handedness of successive stages. With the counter-wound pair, the joint carries its load, admits self-stress at 661 N per cable (the single-handed set admits none), and a passive chain of three such joints stands with less than 3 mm of settling.

Impact attenuation at the bearing-free joint. When the testbed (cables at the specified series-elastic 40 kN/m) is dropped against a mass-matched control whose distal cage is rigid and mounted on a ball joint, the peak ground reaction falls from 287 to 209 N at 0.05 m, from 404 to 282 N at 0.10 m, and from 568 to 388 N at 0.20 m, an attenuation of 27–32% at every height, whereas the hinged humanoid showed an increase (E1). Without a bearing, the elastic network is in series with the load rather than in parallel with a stiff path [3]. The attenuation is due to this series compliance and not to the topology alone: at $10\times$ the cable stiffness (400 kN/m, approaching bare rope) the comparison inverts and the cable joint lands +5 to +21% harder than the pin joint. Geometry and series elasticity are therefore jointly necessary, consistent with bench-scale tensegrity legs [8]. **Prestress does not tune stiffness at rope modulus,** for a material reason that follows from the standard decomposition [23]: the tangent stiffness is the sum of an elastic term of order EA/L and a geometric term of order T/L , and prestress modulates stiffness only when the two are comparable. At the rope modulus of this design, the geometric term is 0.2% of the elastic term; variable stiffness is therefore an actuation cost (co-contraction, E2) rather than an assembly parameter.

The pure tensegrity machine. Generating the entire humanoid as a graph of overlapping cells, without bearings, yields a structure that stands with its foot cells anchored (66 struts, 306 cables; unloaded limb chains sag by approximately 5.9 mm per cell); the pelvis and torso must be rigid strut clusters, and a 22 kg allowance for motors and batteries requires 1000 N of prestress per cable. No controller was attempted for its 210 free DoF, since the 114-actuator cable-driven hinged model already required a planning iteration every 4 ms to stand. This is a limitation of planner tractability rather than controllability; every counter-wound joint has $t_6 > 0$, so the structure is controllable in principle.

VII. WALKING PERFORMANCE OF THE HYBRID

The prescription that follows from the preceding sections, bearings where torque requirements dominate and tensegrity interfaces where compliance dominates, is the hybrid of Sec. III.

Gait synthesis (development configuration, 2.4 kN/m). The controller is the analytic-reference iLQG task with the ankle posture terms removed. With passive ankle cables every configuration tested falls within three seconds, since without ankle actuation there is no center-of-pressure control; welding the feet to the shanks produces immediate walking (10/10 at 0.20 m/s). Providing the planner with tension actuators on two of the four ankle cable families (twelve controls) restores walking at 0.19 m/s in 10/10 trials at a running cost below that of the welded control (0.94 versus 1.19 per step), one family lifting the swing foot and the other pitching the toe. Removing one battery pack (2.6 kg, asymmetric) leaves the gait intact (5/5 at 0.18 m/s).

Design point and stiffness range. The design point is the same walker with every ankle and waist cable at the specified 40 kN/m, run with the development controller without retuning. It walks in 10/10 trials at 0.20 m/s (Wilson 95% lower bound 72%; every trial advanced at least 2.3 m in the 12 s window), with hip and knee duty of 13%, a 99th-percentile command of 36 N m, ankle duty of 13% (122 N) and torso excursion within $\pm 5^\circ$, at $0.034\times$ real time. The walker was also run at 20 and 168 kN/m, each cable retaining its start-pose tension (rest lengths recomputed) and damping ratio. The stiffest cable mode rises from 230 to 664, 939 and 1925 rad/s, so the planner and plant steps are reduced from 8 and 4 ms to keep the product of mode frequency and step at or below the development value, while the 16 ms planning period and all other settings are unchanged (Table IV); a 0.5 ms step would exceed the 512-knot horizon limit of MJPC, truncating the horizon to 0.26 s, in which case every trial fell. Across the range the development controller walks without retuning: 9/10 at 20 kN/m (8/10 upright for the full 16 s) and 5/5 at 168 kN/m, where the nearly rigid interfaces reduce the speed to 0.12 m/s. Leg-torque demand is unchanged (99th percentile 36–37 N m), whereas ankle duty decreases from 18% to 10% and torso excursion from $\pm 10^\circ$ to $\pm 2^\circ$ between the development configuration and 168 kN/m; the development configuration is therefore optimistic with respect to torso

TABLE IV

THE HYBRID WALKER WITH ANKLE AND WAIST CABLES ACROSS THE SPECIFIED SERIES-ELASTIC RANGE UNDER THE DEVELOPMENT CONTROLLER, n TRIALS EACH. THE DESIGN POINT IS IN BOLD; 2.4 kN/m IS THE DEVELOPMENT CONFIGURATION. “WALKS” FOLLOWS THE CRITERION OF SEC. IV; “UPRIGHT” COUNTS TRIALS WITHOUT A FALL IN THE FULL 16 S.

k [kN/m]	steps [ms]	n	walks	upright	m/s	real time
2.4 (development)	8 / 4	10	10/10	10/10	0.19	0.29×
20	2.0 / 1.00	10	9/10	8/10	0.19	0.062×
40 (design point)	1.0 / 0.50	10	10/10	10/10	0.20	0.034×
168	0.8 / 0.41	5	5/5	5/5	0.12	0.030×

compliance and conservative with respect to planning cost (0.062–0.030× real time). *Parameter robustness.* At 20 kN/m, ten trials with every interface cable’s stiffness drawn from $\pm 30\%$ and its prestress from $\pm 20\%$, every body mass from $\pm 15\%$, and ground friction from $\mu \in [0.3, 0.9]$ walk in 10/10 cases at 0.20 m/s, the lowest friction walked being 0.32. Because planner and plant share the perturbed model, this measures the tolerance of the tuning to parameter variation rather than robustness to model mismatch.

Capability envelope (Fig. 4; development configuration unless 40 kN/m is stated). *Speed.* The hybrid has a reliable command band: 6/6 at a commanded 0.30 m/s (0.10 m/s achieved), 6/6 at 0.45 m/s (0.19), falls in every trial at 0.60 m/s, and undertracking below the band. Retuning (cadence 1.1→0.8 Hz, swing-height reference 0.24 m) yields 0.23 m/s in 10/10 trials with 42 mm of clearance (Fig. 2). Speed tracking is approximately 42% of the command against 65–70% for the hinged machine; the remaining panels use the pre-retuning configuration. *Trunk payload.* Boxes on the torso are carried above the tensegrity waist, so that every added kilogram loads the twelve waist cables. The success rate is 10/10 at 2 kg, 9/10 at 5 kg, 3/10 at 8 kg and 0/10 at 12 kg, against an unaffected rigid trunk up to 20 kg; at 12 kg every trial pitches backward within seven seconds, and shifting the load 4 cm forward of the waist axis at a 0.40 m/s command restores 10 kg to 10/10, so the constraint is dynamic pitch balance rather than static capacity. At 40 kN/m ($n = 3$ per mass) the ceiling is unchanged (0/3 at 12 kg), 2 kg walks 3/3, and 5 and 8 kg fall late (0/3, 1/3) at 0.28–0.41 m/s, consistent with the stiffer walker’s closer tracking of the speed command; reducing the command to 0.30 m/s restores 5 kg to 3/3. Payload capacity is therefore set by the controller tuning rather than by cable stiffness. *Arm carry.* Under a common protocol the hybrid holds 2.0 kg at a 12 cm drop threshold against 6.0 kg for the hinged machine, sag-limited versus motor-limited. At 40 kN/m the hybrid’s sag decreases to 8 mm/kg, it holds 4.0 kg, and it too becomes motor-limited (15 N m shoulder saturating at 5 kg); the development configuration’s carry deficit is thus due to chain compliance rather than drive class.

VIII. JOINT-COUNT STUDY AND ALLOCATION RULE

The hybrid occupies one point on an axis from no bearings to a bearing at every DoF. Table V traverses this axis with variants of the same machine, of identical mass and anthro-

TABLE V

THE JOINT-COUNT AXIS. THE MIDDLE ROWS ARE VARIANTS OF THE HYBRID IN ITS DEVELOPMENT CONFIGURATION (2.4 kN/m) UNDER ITS DEVELOPMENT CONTROLLER ($n = 5$); THE ENDPOINTS CARRY THEIR OWN TUNING. “T-JTS” IS THE NUMBER OF BEARING-FREE JOINTS AND “BRGS” THE ARTICULATION HARDWARE MASS (THE HINGED VALUE IS A UNIFORM 70 g/DOF ALLOWANCE).

variant	brgs	T-jts	walks	m/s	note
pure tensegrity	0	all	—	—	stands anchored only
hybrid (design)	0.8 kg	3	10/10	0.19	ankle impact –27..32%
+ rigid waist	0.8 kg	2	0/5 [†]	—	tuning mismatch
+ rigid ankles	0.8 kg	1	5/5	0.21	
+ both rigid	0.8 kg	0	5/5	0.23	
fully hinged (27 DoF)	1.9 kg	0	7/10 [‡]	0.21	impact +14..154%

[†]does not walk under the hybrid’s tuning; with the torso height target returned from 1.20 to 1.28 it walks in 4/5 trials at 0.18 m/s. [‡]own tuning at 0.30 m/s; envelope edge at 0.29 m/s (Fig. 4a).

pometry, in which a rigidized joint retains its mass but loses its tension actuators; the variants run under the pre-retuning controller of the hybrid and are bracketed by the two endpoints with their own tuning.

Bearings versus cable interfaces. A bearing delivers full torque at every joint angle, whereas the authority of a cable joint collapses mid-stride (Table III), and each bearing removes three to five DoF from the planner’s state (46 for the hybrid, 210 for the pure machine). A cable interface, in turn, provides impact attenuation (–27..32% at the ankle testbed), torso compliance ($\pm 5^\circ$ at the design point, $\pm 10^\circ$ in the development configuration), lighter actuation (0.165 versus 0.96 kg drives), graceful degradation, and robustness to mass changes (battery-removal test). Its costs are reduced *speed* (rigidizing increases gait speed monotonically, 0.19 → 0.21 → 0.23 m/s under one controller in the development configuration), increased *planner load* (0.034× against 0.60× real time), reduced *payload* (Fig. 4b), and *tuning sensitivity*: the gait tuned for the compliant waist does not walk (0/5) when that joint is rigidized. Success rates along the axis are indistinguishable at these sample sizes; the speed ordering is the measured effect.

Allocation rule. Bearings should be placed where the gait’s torque requirements concentrate, and series-elastic cable interfaces where impacts enter the structure. The hips and knees carry the largest, most pose-dependent torques and no first impact; the ankles contact the ground first and require substantially less moment (roughly half at the 99th percentile); the waist is the marginal case, providing torso compliance at a cost in speed, payload and tuning. The rule reflects present-day drives and planners rather than a fundamental limit.

IX. DISCUSSION

Limitations. All results are obtained in simulation, without tendon friction, hysteresis, creep or cable mass. The headline walker, payload, carry, robustness and joint drop tests are evaluated at the specified 20–168 kN/m; the speed sweep, joint-count study and ankle controls remain at the 2.4 kN/m development configuration. In hardware, tendon friction and hysteresis will bias the delivered tension and cause prestress to drift; the series-elastic element mitigates both by converting

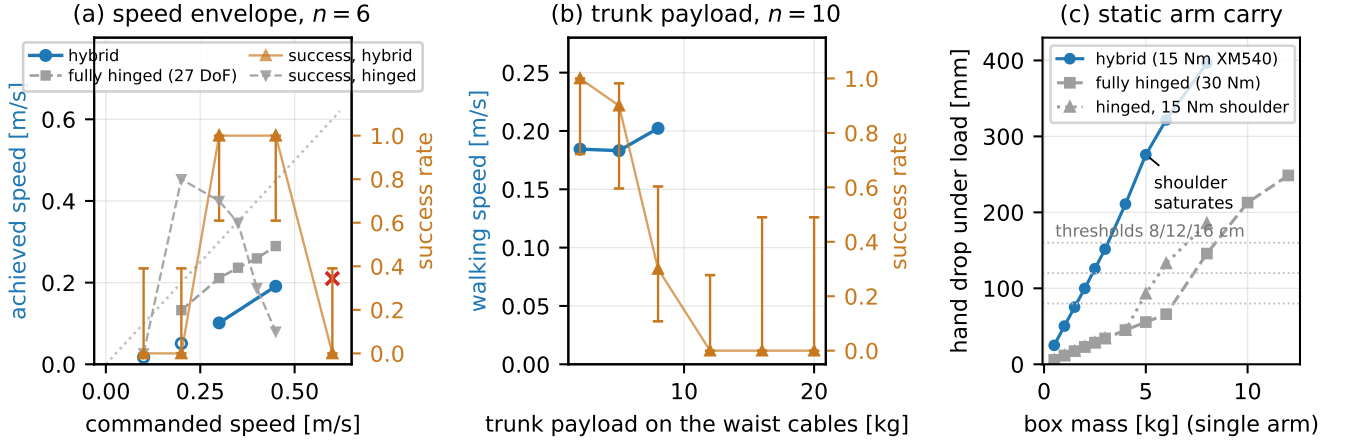


Fig. 4. Capability sweeps of the hybrid in the development configuration (interface cables at 2.4 kN/m; results at 40 kN/m are given in the text). Error bars are Wilson 95% intervals on the success rate. (a) Achieved speed (left axis) and success rate (right axis) versus commanded speed for the hybrid ($n = 6$, orange) and the fully hinged machine (gray, $n = 10$, scored under its own criterion); 1:1 line dotted. Open circles mark hybrid commands (0.10–0.20 m/s) that produce upright undertracking rather than falls; the cross marks falls at 0.60 m/s. (b) Trunk payload above the tensegrity waist ($n = 10$ up to 12 kg, $n = 4$ beyond); the rigid trunk of the hinged machine was unaffected up to 20 kg (own success criterion). (c) Static single-arm carry under a common protocol (pelvis welded, identical pose, driven hinges PD-held): hand drop versus box mass, thresholds dotted; the hybrid is sag-limited (53 mm/kg), the hinged machine motor-limited (16 mm/kg until its 30 N m shoulder saturates), and halving that shoulder class (dotted) changes the hinged carry only marginally.

tension into a displacement measurement, but tension sensing and periodic re-tensioning must be provided for, while cable mass (0.10 kg) is negligible. The planner runs at $0.034\times$ real time at the design point; a shorter horizon on a reduced model, a multi-rate scheme with a fast tension loop, or a policy distilled from the planner’s rollouts would be required to close this gap, none of which is evaluated here. The AK70-10-class hip and knee drives (24.8 N m peak, approximately 8 N m rated) deliver approximately 77 N m peak and 26 N m continuous through the 3:1 tendon reduction; the planner’s 80 N m bound corresponds to the peak, the mean command (13% duty) lies within the continuous rating, and the 99th percentile (36 N m) within the peak. Over 120 s of logged walking at the design point, commands above the 74 N m drive peak occur in 2 isolated episodes of at most 1.5 ms, commands above the continuous rating occupy 4.4% of the time, and the RMS demand, which determines thermal load, is 5.3 N m at the motor for the worst joint, 67% of the rating. The sample sizes ($n = 3$ –10) resolve speed differences but not success-rate differences of one or two trials.

X. CONCLUSION

Within a single design family, a humanoid composed only of tensegrity cells stands when anchored, given overlapping cages and counter-wound cable families, but presents 210 free DoF for which no controller is available; a fully hinged variant walks, but its cables increase impact loads and cannot deliver the specified torques at gait poses. Between these extremes, a 1.68 m, 20.1 kg hybrid with bearings at the eight torque-carrying DoF and tensegrity joints where impacts enter walks at 0.20 m/s in 10/10 trials at the specified cable stiffness, with a gait that transfers across the entire 20–168 kN/m range. Rigidizing its tensegrity joints increases speed and removes

compliance; removing its bearings removes planner tractability. As a design study, the results indicate that the concept warrants hardware development: the machine has a third to two thirds of the mass of size-matched humanoids and is controllable by a predictive planner. The allocation rule and trade-off data constitute one iteration of an MPC-in-the-loop design process and identify the parameters for the next, namely moment-arm geometry and cable stiffness.

REPRODUCIBILITY AND ACKNOWLEDGMENT

All code (generator, MJPC task, experiments and analysis) is available in anonymized form at <https://anonymous.4open.science/r/tensegrity-7C62/>; the accompanying video shows the gait, and a single script regenerates every figure and quantity from the logged runs. AI has been used to support all aspects of this work (Claude Fable 5.1, Anthropic, via Claude Code). Specifically, (a) *code*: the parametric MuJoCo model generator, the MJPC task and analytic gait reference (Sec. IV), the experiment, dimensioning and analysis scripts (Secs. III–VIII), and the pipeline that regenerates every reported quantity were written by the AI to the authors’ specifications and corrected under their direction; (b) *figures*: Figs. 1, 2 and 4 were rendered or plotted by AI-written code from logged data, and Fig. 3 was drawn by the AI; (c) *text*: the manuscript, including the abstract and all sections, was drafted by the AI from the authors’ design decisions, results and outline, and was reviewed, corrected and edited by the authors, who defined the research question and design goals, directed every experiment, verified the results against the simulation logs, and take full responsibility for the content.

REFERENCES

- [1] R. B. Fuller, “Tensile-integrity structures,” U.S. Patent 3,063,521, 1962.

- [2] K. D. Snelson, "Continuous tension, discontinuous compression structures," U.S. Patent 3,169,611, 1965.
- [3] A. P. Sabelhaus, J. Bruce, K. Caluwaerts, P. Manovi, R. F. Firoozi, S. Dobi, A. M. Agogino, and V. SunSpiral, "System design and locomotion of superball, an untethered tensegrity robot," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 2867–2873.
- [4] K. Caluwaerts, J. Despraz, A. Iscen, A. P. Sabelhaus, J. Bruce, B. Schrauwen, and V. SunSpiral, "Design and control of compliant tensegrity robots through simulation and hardware validation," *Journal of The Royal Society Interface*, vol. 11, no. 98, p. 20140520, 2014.
- [5] D. S. Shah, J. W. Booth, R. L. Baines, K. Wang, M. Vespignani, K. Bekris, and R. Kramer-Bottiglio, "Tensegrity robotics," *Soft Robotics*, vol. 9, no. 4, pp. 639–656, 2022.
- [6] E. Jung, V. Ly, N. Cessna, M. L. Ngo, D. Castro, V. SunSpiral, and M. Teodorescu, "Bio-inspired tensegrity flexural joints," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 5561–5566.
- [7] Q. Wen, M. Zhang, J. Sun, W. Li, J. Chu, Z. Wang, S. Zhang, and L. Ren, "Legged robot with tensegrity feature bionic knee joint," *Advanced Science*, vol. 12, no. 12, p. 2411351, 2025.
- [8] E. Mortensen, J. Petrus, A. Dittrich, and D. Floreano, "Tensegrity-based robot leg design with variable stiffness," *arXiv preprint arXiv:2504.19685*, 2025.
- [9] A. R. Ali and H. S. Abdullah, "Development of a compliant spine mechanism for enhanced humanoid robotics locomotion," *Scientific Reports*, vol. 15, p. 44646, 2025.
- [10] W. Li and E. Todorov, "Iterative linear quadratic regulator design for nonlinear biological movement systems," in *1st International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, 2004, pp. 222–229.
- [11] Y. Tassa, T. Erez, and E. Todorov, "Synthesis and stabilization of complex behaviors through online trajectory optimization," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 4906–4913.
- [12] T. Howell, N. Gileadi, S. Tunyasuvunakool, K. Zakka, T. Erez, and Y. Tassa, "Predictive sampling: Real-time behaviour synthesis with MuJoCo," *arXiv preprint arXiv:2212.00541*, 2022.
- [13] C. R. Calladine, "Buckminster fuller's "tensegrity" structures and clerk maxwell's rules for the construction of stiff frames," *International Journal of Solids and Structures*, vol. 14, no. 2, pp. 161–172, 1978.
- [14] P. Bosscher and I. Ebert-Uphoff, "Wrench-based analysis of cable-driven robots," in *2004 IEEE International Conference on Robotics and Automation (ICRA)*, 2004, pp. 4950–4955.
- [15] J. M. Friesen, J. L. Dean, T. Bewley, and V. SunSpiral, "A tensegrity-inspired compliant 3-DOF compliant joint," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 3301–3306.
- [16] M. Gouttefarde, J.-P. Merlet, and D. Daney, "Wrench-feasible workspace of parallel cable-driven mechanisms," in *2007 IEEE International Conference on Robotics and Automation (ICRA)*, 2007, pp. 1492–1497.
- [17] R. E. Skelton and M. C. de Oliveira, *Tensegrity Systems*. Springer, 2009.
- [18] A. P. Sabelhaus, H. Zhao, E. L. Zhu, A. K. Agogino, and A. M. Agogino, "Model-predictive control with inverse statics optimization for tensegrity spine robots," *arXiv preprint arXiv:1806.08868*, 2018.
- [19] A. P. Sabelhaus, H. Ji, P. Hylton, Y. Madaan, C. Yang, A. M. Agogino, J. Friesen, and V. SunSpiral, "Mechanism design and simulation of the ULTRA Spine: A tensegrity robot," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE)*, 2015.
- [20] A. P. Sabelhaus, A. K. Akella, Z. A. Ahmad, and V. SunSpiral, "Design, simulation, and testing of a flexible actuated spine for quadruped robots," *arXiv preprint arXiv:1804.06527*, 2018.
- [21] S. Lessard, D. Castro, W. Asper, S. D. Chopra, L. B. Baltaxe-Admony, M. Teodorescu, V. SunSpiral, and A. Agogino, "A bio-inspired tensegrity manipulator with multi-dof, structurally compliant joints," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 5515–5520.
- [22] D. Zappetti, R. Arandes, E. Ajanic, and D. Floreano, "Variable-stiffness tensegrity spine," *Smart Materials and Structures*, vol. 29, no. 7, p. 075013, 2020.
- [23] S. D. Guest, "The stiffness of prestressed frameworks: A unifying approach," *International Journal of Solids and Structures*, vol. 43, no. 3–4, pp. 842–854, 2006.
- [24] A. G. Tibert and S. Pellegrino, "Review of form-finding methods for tensegrity structures," *International Journal of Space Structures*, vol. 18, no. 4, pp. 209–223, 2003.
- [25] Y. Nakanishi, Y. Asano, T. Kozuki, H. Mizoguchi, Y. Motegi, M. Osada, T. Shirai, J. Urata, K. Okada, and M. Inaba, "Design of upper limb by adhesion of muscles and bones: Detail human mimetic musculoskeletal humanoid Kenshiro," in *2013 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2013.
- [26] Y. Asano, K. Okada, and M. Inaba, "Design principles of a human mimetic humanoid: Humanoid platform to study human intelligence and internal body system," *Science Robotics*, vol. 2, no. 13, p. eaaq0899, 2017.
- [27] G. A. Pratt and M. M. Williamson, "Series elastic actuators," in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 1995, pp. 399–406.
- [28] N. G. Tsagarakis, S. Morfe, G. Medrano-Cerda, Z. Li, and D. G. Caldwell, "COMpliant huMANoid COMAN: Optimal joint stiffness tuning for modal frequency control," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2013, pp. 673–678.
- [29] B. T. Quinlivan, S. Lee, P. Malcolm, D. M. Rossi, M. Grimmer, C. Sivi, N. Karavas, D. Wagner, A. Asbeck, I. Galiana, and C. J. Walsh, "Assistance magnitude versus metabolic cost reductions for a tethered multiarticular soft exosuit," *Science Robotics*, vol. 2, no. 2, p. eaah4416, 2017.
- [30] S. Ha, S. Coros, A. Alspach, J. Kim, and K. Yamane, "Computational co-optimization of design parameters and motion trajectories for robotic systems," *The International Journal of Robotics Research*, vol. 37, no. 13–14, pp. 1521–1536, 2018.
- [31] C. Schaff, D. Yunis, A. Chakrabarti, and M. R. Walter, "Jointly learning to construct and control agents using deep reinforcement learning," in *2019 IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 9798–9805.
- [32] Y. Tassa, N. Mansard, and E. Todorov, "Control-limited differential dynamic programming," in *2014 IEEE International Conference on Robotics and Automation (ICRA)*, 2014, pp. 1168–1175.
- [33] J. Koenemann, A. Del Prete, Y. Tassa, E. Todorov, O. Stasse, M. Bennewitz, and N. Mansard, "Whole-body model-predictive control applied to the HRP-2 humanoid," in *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2015, pp. 3346–3351.
- [34] E. Dantec, M. Naveau, P. Fernbach, N. Villa, G. Saurel, O. Stasse, M. Taix, and N. Mansard, "Whole-body model predictive control for biped locomotion on a torque-controlled humanoid robot," in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, 2022, pp. 638–644.
- [35] M. H. Raibert, *Legged Robots That Balance*. MIT Press, 1986.
- [36] J. Pratt, J. Carff, S. Drakunov, and A. Goswami, "Capture point: A step toward humanoid push recovery," in *2006 6th IEEE-RAS International Conference on Humanoid Robots*, 2006, pp. 200–207.
- [37] D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th ed. John Wiley & Sons, 2009.
- [38] 1X Technologies, "NEO gamma: technical specifications," Manufacturer specification, 2025, 165 cm, 30 kg, tendon-driven joints, soft covers.
- [39] K. Kaneko, F. Kanehiro, M. Morisawa, K. Akachi, G. Miyamori, A. Hayashi, and N. Kanehira, "Humanoid robot HRP-4: Humanoid robotics platform with lightweight and slim body," in *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2011, pp. 4400–4407.
- [40] Unitree Robotics, "G1 humanoid robot: technical specifications," Manufacturer specification, 2024, 132 cm, 35 kg, quasi-direct-drive joints.
- [41] Fourier Intelligence, "GR-1 humanoid robot: technical specifications," Manufacturer specification, 2023, 165 cm, 55 kg, 40 DoF, 300 N m peak hip torque.
- [42] PAL Robotics, "REEM-C biped humanoid research robot: technical specifications," Manufacturer specification, 2013, 165 cm, 80 kg, 44 DoF.
- [43] G. Scarr, *Biotensegrity: The Structural Basis of Life*. Handspring Publishing, 2014.
- [44] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.